

AI Brand Identity & Naming Generator – Hackathon Problem Brief

1. Problem Context

Early-stage entrepreneurs and solopreneurs have a great business idea but face a creative bottleneck: finding a memorable brand name, crafting a tagline, and defining visual identity (colors, fonts, logo direction) requires hiring expensive branding consultants or spending weeks researching on their own. They manually brainstorm names, check domain availability by hand, and piece together mood boards from Pinterest. Existing tools like Namelix or Brandmark handle naming or logo generation separately, but there's no unified, AI-powered experience that generates names, taglines, visual direction, and logo concepts all in one session. Early-stage founders need speed to validate their brand vision before investing in full rebrand design work.

2. Objective

Build a web application where users describe their business idea, and an AI instantly generates multiple brand name options, matching taglines, a visual direction (color palette and font pairing suggestions), and AI-generated logo concept sketches. Include domain availability checks so users can immediately assess whether their ideal name is actionable.

3. Core Features (Must-Have)

- **Business Idea Input Form:** Capture details—business/product name (if any), industry, target audience, key values/mission, tone (e.g., playful, professional, minimalist, bold), and any specific requirements ("must sound tech-forward," "must be short," "avoid animal names").
- **AI Brand Name Generation:** Call an AI API (OpenAI) with the business description to generate 10–15 creative brand name options that fit the industry, tone, and constraints. Names should be diverse (punny, descriptive, invented words, etc.).
- **Tagline Generation:** For each brand name, generate a matching short tagline (5–10 words max) that reinforces the brand promise or mission.
- **Domain & Social Handle Availability Check:** For each generated name, check if the .com domain is available using a WHOIS API or domain lookup service. Also suggest .io, .co, and check if the name is available on Twitter/Instagram (mock this if APIs are expensive).
- **Visual Direction Suggestions:** Generate a visual brand guide for each name option—recommended color palette (3–5 hex colors with names), font pairing (headline font + body font suggestions), and a brief style description (e.g., "Modern and clean with warm accents").
- **AI Logo Concept Generation:** For the user's selected name, generate 3–5 AI logo concept sketches using DALL-E or Stability AI that reflect the brand's vibe and visual direction.
- **Brand Identity Card Export:** Create a downloadable PDF or image card showing the selected brand name, tagline, color palette, fonts, logo concepts, and domain availability summary—a one-pager founders can share with investors or co-founders.

4. Stretch Features (Good-to-Have)

- **Interactive Brand Selection & Refinement:** Allow users to click "Generate alternatives" for a specific name to see more taglines or logo concepts without regenerating the entire set.
- **Mood Board Generator:** Curate a Pinterest-style visual mood board (using Unsplash API) that matches the selected brand's visual direction, showing design inspiration.
- **Social Media Preview:** Show mockups of how the brand name and logo would look as a Twitter/LinkedIn header or Instagram profile.
- **Brand Guidelines Template:** Generate a skeleton brand guidelines document (PDF) with logo usage rules, color specs, typography hierarchy, and tone-of-voice guidelines that founders can flesh out later.
- **Competitor Brand Comparison:** Given the selected brand name, fetch and display 2–3 competitor brands in the same space so users can assess differentiation.

5. Suggested Tech Stack

Frontend: React.js with TypeScript – Build an intuitive multi-step form (input → name results → visual direction → logo selection → export). React's component model makes it easy to display and switch between brand options.

Backend: Node.js + Express – Handle API orchestration (calls to OpenAI for naming, domain checking, logo generation), manage file exports, and store user sessions.

Database: Firebase Firestore or MongoDB Atlas – Store user projects, generated brand options, and favorites. Firestore is good for quick prototyping and includes built-in auth.

AI for Naming & Taglines: OpenAI GPT-3.5 or GPT-4 – Excellent at generating creative names and taglines that fit specific constraints and tones. Use function calling to structure output as JSON for easy parsing.

AI for Logo Generation: DALL-E 3 API or Stability AI / Replicate API – DALL-E is more refined but pricier (~\$0.08 per image); Stability/Replicate offers cheaper Stable Diffusion access. For hackathon, consider pre-generating a few logo styles and reusing them to save cost.

Domain Availability Check:

- **Whois API** (<https://whois.api.com/>) or **Namecheap API** – Checks .com availability programmatically.
- **Hunter.io API** or **RapidAPI Domain Check** – Lightweight alternatives for domain lookups.
- **Mock Service** – For hackathon speed, hardcode a list of "available" and "taken" domains to simulate checks without hitting rate limits.

Social Handle Checker:

- **Twitter/X API** (rate-limited, requires approval) or **Bluesky API** – Can check handle availability.
- **Mock Service** – Simulate checks by returning random availability status during hackathon.

PDF Export: jsPDF + html2canvas – Generate downloadable brand identity cards without backend complexity.

Deployment: Vercel (frontend) + Render/Railway (backend) – Free tiers, instant git deployments.

6. Suggested Approach / Step-by-Step Plan

1. **Project Setup (20 min):** Initialize React frontend, Node/Express backend, Firebase project, and set up environment variables for OpenAI and domain check APIs.
2. **Design Database Schema (15 min):** Plan collections: Users, BrandProjects (stores input details), GeneratedNames (stores AI-generated names, taglines, colors, fonts), LogoConcepts (stores image URLs). Keep it simple.
3. **Build Business Input Form (30 min):** Create a multi-field form capturing industry, target audience, tone, values, and constraints. Save to database on submission. No AI yet.
4. **Implement Naming API Endpoint (45 min):** Write `/generate-names` backend endpoint. Take business details, construct a detailed OpenAI prompt (e.g., "Generate 12 brand names for a sustainable fashion startup targeting eco-conscious millennials. Names should be modern, memorable, and under 12 characters. Include a mix of descriptive, invented, and playful names. Return as JSON array with fields: name, meaning."), call OpenAI, parse response into JSON, and return to frontend.
5. **Build Tagline Generation (30 min):** Create `/generate-taglines` endpoint that takes brand names and business description, calls OpenAI to generate matching taglines (1 per name), and returns structured data.
6. **Implement Domain Availability Check (30 min):** Write `/check-domains` endpoint that takes brand names and pings a WHOIS API (or mocks it) to check `.com` availability. Return availability status for each name.
7. **Build Visual Direction API (30 min):** Create `/generate-visual-direction` endpoint that takes a brand name and business vibe, calls OpenAI to suggest a color palette (return as hex codes with names) and font pairing (headline + body), formatted as JSON.
8. **Implement Logo Generation (45 min):** Build `/generate-logos` endpoint that takes a brand name, visual direction, and vibe, calls DALL-E or Stable Diffusion API to generate 3–5 logo concepts. Store image URLs and display them. For cost control, cache logos and reuse for demo.
9. **Build Results Display & Selection UI (1 hour):** Create a React component that shows generated names in a grid/list format, displays taglines, visual palettes, and domain availability. Let users click to select/favorite options and regenerate specific items (names, taglines, logos) without redoing the whole project.

10. **Implement Export & Polish (45 min):** Build a branded PDF/image export showing selected name, tagline, colors, fonts, logos, and domain info. Test end-to-end, deploy, and prepare demo script.

7. Sample Data / APIs to Use

OpenAI API (<https://platform.openai.com/>) – For naming and tagline generation. Example prompt:

```
"A user is building a [INDUSTRY] startup targeting [AUDIENCE].  
Their mission is: [MISSION].  
Brand tone: [TONE]. Constraints: [CONSTRAINTS].  
Generate 12 creative brand names that fit these criteria. For  
each name, provide:  
1. The name (max 12 characters)  
2. A one-line meaning/explanation  
Format as JSON array with keys: name, meaning."
```

DALL-E 3 API (<https://platform.openai.com/docs/guides/images>) – For logo generation. Example prompt:

```
"Create a logo concept for a brand called '[BRAND_NAME]'. The  
brand is in [INDUSTRY], targets [AUDIENCE],  
and has this visual direction: colors [COLOR_PALETTE], fonts  
[FONT_PAIRING], style [STYLE_DESCRIPTION].  
The logo should be modern, minimalist, and work on both white  
and dark backgrounds.  
Generate one cohesive logo concept that embodies the brand's  
identity."
```

Whois API / Domain Checker (<https://whois.api.com/> or via Namecheap) – Check .com, .io, .co availability for generated names.

Unsplash API (<https://unsplash.com/developers>) – Optional: fetch color-matched reference images for the mood board (stretch feature).

Mock Data for Testing:

- Pre-create 5 sample brand projects with generated names, taglines, colors, fonts, and logo image URLs.
- Hardcode domain availability (80% available, 20% taken) to simulate checker without API calls.
- Pre-generate 10–15 sample logo images using Figma or Unsplash, tag them by style (modern, playful, corporate, minimal), and reuse during demo.

8. Key Challenges to Anticipate

Challenge 1: Brand Names Generated by AI Can Be Generic or Unmemorable

GPT can produce technically correct names, but they sometimes lack personality or memorability. A name that sounds good on paper might not feel right as a brand. **Tip:** Refine your OpenAI prompt with specific examples of good brand names in the user's industry (e.g., "like Spotify, Canva, Notion—modern, one-word, memorable"). Ask the model to rank names by memorability. Let users regenerate specific batches of names (e.g., "more playful names" or "more serious names") without regenerating taglines.

Challenge 2: Domain Checker Rate Limits and Costs

WHOIS APIs have rate limits and can be expensive. Checking 12 domains $\times$ 3 TLDs = 36 API calls per user can blow through free tiers. **Tip:** Batch all domain checks into a single endpoint call, cache results, and mock availability during development. For the hackathon demo, pre-check 5–10 sample names and reuse their results. Consider only checking .com (most important) if hitting limits.

Challenge 3: Image Generation (DALL-E) is Slow and Expensive

Generating 5 logo concepts per user takes 30–60 seconds and costs ~\$0.40–\$0.50. Users might request regenerations, ballooning cost. **Tip:** Generate logos asynchronously after the user selects a final brand name, not for every name upfront. Show a loading state and set expectations ("Generating your logo concepts—this takes ~30 seconds"). For the demo, pre-generate 2–3 logo sets and cache them; reuse the same logos for multiple demo runs to save money.

Challenge 4: UI Complexity with Multiple Generations and Selections

Managing state for 12 names $\times$ taglines $\times$ colors $\times$ fonts $\times$ logos can get messy. Users might want to mix-and-match (name from option 3, colors from option 7, logo from option 5). **Tip:** Use React state management (useState or Context) to track selections cleanly. Display all options in a grid/card layout with a "select this combination" button, or allow individual selections with a "custom combo" mode. Keep the UI simple for the hackathon—stick to sequential selections (pick name $\rightarrow$ pick colors $\rightarrow$ pick logo).

9. What a Good Demo Looks Like

A compelling demo shows a judge filling out a quick form: "Building a sustainable productivity app for remote teams, modern and eco-conscious vibe, must feel trustworthy." The app instantly displays 12 brand name options (e.g., "Watershed," "Drift," "Compass," "Bloom") in a grid, each with a tagline and domain availability status (✓ available, ✗ taken). The judge clicks on a name, and the app reveals a color palette (e.g., sage green, cream, deep blue), font recommendations (e.g., "Inter + Merriweather"), and a style description. The judge clicks "Generate Logo Concepts," and within 30 seconds, 4 logo sketches appear—modern, minimal, brand-appropriate. The judge selects their favorite combination and clicks "Export Brand Identity." A PDF downloads showing the selected name, tagline, color hex codes, fonts, logo concepts, and domain availability—ready to share with co-founders or investors. The entire flow takes under 2 minutes, feels intuitive, and generates genuinely usable brand direction that doesn't require design experience.